SkyAgent: Revisable Target Memory for Continuous Aerial Search

Kaibo Yang[1,2,3], Xingyu Wu[1,2,3], Enze Zhu[1,2,3], Yajie Chai[4], Xuyang Li[1,2,3,4], Jiahui Cui[1,2,3], Xiaoxuan Liu[1,2], Lei Wang[1,2], Zhan Chen[1,2]*

[1]Aerospace Information Research Institute, Chinese Academy of Sciences
[2]Key Laboratory of Target Cognition and Application Technology (TCAT)
[3]School of Electronic, Electrical and Communication Engineering, University of Chinese Academy of Sciences
[4]Baidu Inc.

Abstract

Continuous aerial search builds target hypotheses from sparse, viewpoint-dependent observations. When later evidence contradicts a stored hypothesis, however, the candidate can remain available to the planner and continue triggering approach or verification actions. We call this trajectory-level failure *false candidate persistence* (FCP). We equip SkyAgent with *Revisable Target Memory* (RTM), which associates evidence with persistent candidates, discounts correlated views, and revises each candidate through Tentative, Confirmed, and Rejected states. The key interface binds every generated waypoint to its source candidate: rejection therefore invalidates stale goals, while conservative reactivation protects temporarily occluded targets. Across four splits from the CityAVOS, UAV-ON, and OpenUAV benchmarks, our SkyAgent equipped with RTM outperforms all competitors, improving SR by 2.90–6.31 points while substantially reducing Behavior-FCP and post-contradiction travel with a low false-rejection rate. Across 35 matched real-world DJI scenarios per method, RTM consistently improves task success and behavioral reliability, further validating its state-to-action correction under physical deployment conditions.

1 Introduction

In a representative aerial search, a distant rectangular region is logged as a target candidate. A later side-angle view reveals that the region is a facade reflection, not the target. This revision problem is central to continuous aerial object search: the physical target is static, but evidence about each candidate shifts as the vehicle approaches, changes viewpoint, or encounters occlusion (Ji et al. 2026; Xiao et al. 2025; Wang et al. 2025). The agent rarely sees decisive evidence in a single frame; a distant shape, a brief gap in foliage, or an intermittent specular return may be all that a long trajectory provides.

Existing aerial agents have made progress on two fronts. *Memory-based* methods retain observations across frames through asynchronous maps, octree representations, or filtered write-in (Zhang et al. 2026a; Zhou et al. 2026; Ji et al. 2026). *Verification-based* methods seek additional viewpoints before committing to a stop or reconsider a selected snapshot before executing it (Jang and Kim 2026; Cao et al. 2025; Ren et al. 2024; Wang et al. 2026a). Both fronts focus on the *forward* direction: building or validating confidence before a decision is made. We focus on what happens *after*:

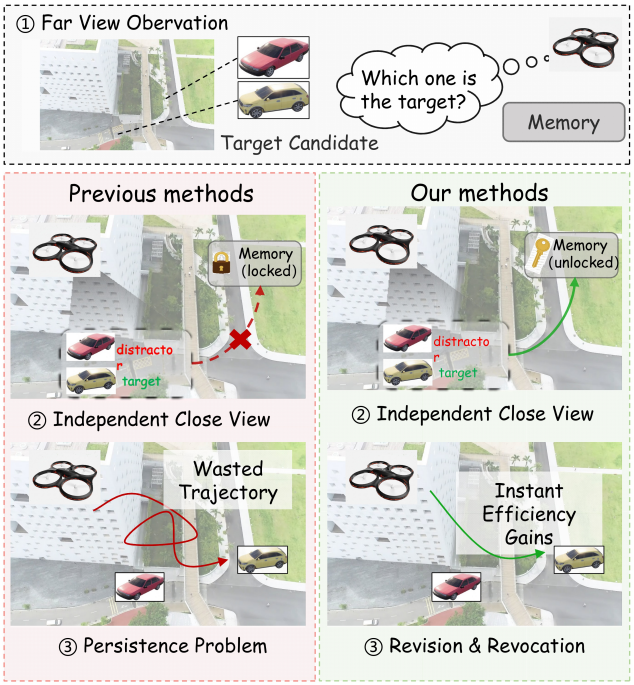


Figure 1: **False candidate persistence in continuous aerial search.** A static distractor looks like the target from an early viewpoint (*A*) and is contradicted by an independent later view (*B*). An append-only state keeps C_1 active; the drone wastes route on a known-wrong candidate (*C*, top). RTM rejects C_1, cancels its pending waypoint, and resumes search toward the true target without erasing the audit trail (*C*, bottom).

when a later view contradicts a candidate already sitting in persistent memory, can the system actually change that memory and cancel the goals it generated? In the systems we study, the answer is often no—the candidate remains available to planning even after a closer view proves it wrong.

We define *false candidate persistence* (FCP) as a trajectory-level failure: after a candidate receives sufficient reference contradiction, it stays in the active plan (State-FCP) or again induces approach, revisit, or confirmation actions (Behavior-FCP). FCP is not a transient detector error, which

*Corresponding author.

is a wrong observation in a single frame; it is a known-wrong hypothesis that continues to consume distance, time, and planning cycles through stale memory state. Figure 1 illustrates this failure and its correction.

The fix is not more storage. It is different memory semantics: candidates should be revisable hypotheses, not committed facts. A candidate must retain both supporting and opposing evidence, discount redundant adjacent frames, and expose an explicit lifecycle state to the planner. Changing a score alone is not enough; when a candidate is rejected, every pending waypoint it sourced must be invalidated before the next planning cycle. At the same time, rejection must not be permanent, because temporary occlusion and low-quality frames can produce misleading negative evidence.

We equip the SkyAgent execution backbone with *Revisable Target Memory* (RTM). *Candidate Evidence Memory* associates observations with persistent candidates, groups them by independent viewpoint, and accumulates signed evidence. *Revise-and-Suppress* maps accumulated evidence to lifecycle states through a state-dependent transition function and cancels all stale waypoints on rejection. The existing fast search policy, event trigger, inspection skills, and safety-projected controller remain as execution infrastructure; candidate truth state is managed exclusively through RTM.

Our contributions are three: (1) we identify false candidate persistence (FCP) as a trajectory-level failure distinct from transient detection errors, and introduce complementary state-, behavior-, and cost-level measures that separate a stored false hypothesis from one that still affects flight; (2) we propose Revisable Target Memory (RTM), which couples candidate lifecycle revision to planner binding: rejecting a candidate invalidates every unexecuted goal it sourced, while conservative reactivation permits recovery after new independent positive evidence; (3) we evaluate RTM through natural FCP prevalence, controlled comparisons and mechanism ablations on three aerial benchmarks, and 35 matched DJI scenarios per method (Table 1).

2 Related Work

Aerial target search and persistent memory. Classical aerial target-search systems improve coverage through visibility-aware viewpoint selection, hierarchical planning, probabilistic target beliefs, or sparse history maps (Luo et al. 2024; Ray et al. 2024; Hou et al. 2026). Semantic aerial navigation has since moved from instruction-following benchmarks such as AerialVLN, OpenUAV, and CityNav (Liu et al. 2023; Wang et al. 2025; Lee et al. 2025) to city-scale planners with topo-metric, annotated-map, hierarchical-memory, and VLM reasoning modules (Gao et al. 2024; Zhang et al. 2025a,b; Cai et al. 2025). Training-free aerial frameworks include SPF's image-space waypoint grounding and PSC-AVDN's parsing–search–confirmation with structured spatial memory; HTNav and LookasideVLN instead learn tiered or direction-aware policies (Hu et al. 2025; Qi et al. 2026; Fan et al. 2026; Ning et al. 2026). The adjacent aerial object-goal line introduces open-world target descriptions and long-horizon search in UAV-ON and CityAVOS (Xiao et al. 2025; Ji et al. 2026), with APEX, OctMem-Agent, NavFoM, and RAVEN scaling retrieval through asynchronous maps, octrees, foundation-model memory, or voxel–ray representations (Zhang et al. 2026a; Zhou et al. 2026; Zhang et al. 2026b; Kim et al. 2025).

Persistent memory is also central to ground ObjectNav and lifelong navigation. VLMaps and ConceptGraphs attach open-vocabulary semantics to metric maps or 3D scene graphs (Huang et al. 2023; Gu et al. 2024); MemoNav, 3D-Mem, and AstraNav-Memory compress observations into working, snapshot, or image-centric long-term memories (Li et al. 2024; Yang et al. 2025; Hu et al. 2026). GOAT-Bench explicitly evaluates memory across a sequence of heterogeneous goals, while TrajRAG retrieves geometric–semantic experience accumulated across episodes (Khanna et al. 2024; Wang et al. 2026b). These methods chiefly improve representation, retention, retrieval, or coverage. RTM addresses a downstream failure mode: after new evidence contradicts a stored candidate, whether that candidate remains available to the planner and what happens to every goal it already generated.

Revision and verification. Active perception acquires informative views before committing to a decision (Bajcsy 1988); APRR applies this principle to refine the last-mile viewpoint for precise ObjectNav (Qin et al. 2025). Recent navigation agents verify different objects of uncertainty. PSC-AVDN confirms a candidate region after a structured search (Qi et al. 2026); Context-Nav scores memory-conditioned viewpoints, CogNav models a cognitive action process, and SCOPE revisits promising locations before committing (Jang and Kim 2026; Cao et al. 2025; Wang et al. 2026a). UniGoal uses graph matching for goal-conditioned aerial decisions, while WMNav corrects short-horizon actions through a learned world model (Yin et al. 2025; Nie et al. 2025). Explore-Until-Confident calibrates stopping in embodied question answering rather than target navigation (Ren et al. 2024). CARe is especially close: it combines confidence and multi-view consistency to replan decisions induced by an erroneous pre-explored semantic map (Ko et al. 2025). Classical multiple-hypothesis tracking provides a related probabilistic view of preserving alternatives under ambiguous association (Blackman 2004). RTM does not claim candidate lifecycles or replanning in isolation. Its distinction is a candidate-local revision contract: every waypoint records its source candidate, and rejection invalidates all unexecuted goals sourced from that candidate rather than only reconsidering the next action.

Online agent memory correction. VLM self-correction can revise semantic grounding errors under targeted prompts (Liao et al. 2025), while Reflexion converts task feedback into linguistic episodic memory for later trials (Shinn et al. 2023). Neither mechanism specifies how a continuously acting navigation system should retract the downstream commitments of a disproved perceptual hypothesis. RTM instead operates online and candidate-locally: it retains supporting and contradicting observations for audit, changes a candidate's availability to planning during the same episode, cancels its still-pending waypoints, and permits recovery only after independent positive views arrive.

3 Problem Setup

Task. Single-frame object detection assumes each observation is independent. Continuous aerial search instead builds persistent candidate hypotheses from a stream of partial, viewpoint-dependent observations. We therefore define the candidate representation to support evidence accumulation and lifecycle revision. At each step t, the agent receives an observation o_t, and the perception front end produces an evidence item e_t:

$$o_t = (I_t,\, p_t,\, q),$$
$$e_t = (R_t,\, \hat{x}_t,\, f_t,\, y_t,\, \rho_t).$$

Here I_t is the current image, p_t is the vehicle pose, and q is an image or semantic target query. The candidate region R_t contains a crop or mask, estimated 3-D location $\hat{x}_t$, multimodal feature f_t, signed query-relative evidence $y_t \in [-1, 1]$, and quality $\rho_t \in [0, 1]$. Thus every evidence item refers to one localized hypothesis rather than assigning an image-level score to multiple candidates. We study physically static targets throughout. A candidate's spatial location does not change, but its identity judgment and associated belief score are updated as new evidence arrives.

Candidate state. A binary active/inactive flag is insufficient because approach, final verification, and complete removal require different planner permissions. The i-th persistent candidate is therefore represented as

$$m_i^t = \big(x_i,\, \mathcal{E}_i^{+,t},\, \mathcal{E}_i^{-,t},\, b_i^t,\, z_i^t,\, n_i^{+,\mathrm{new}}\big), \tag{1}$$

where x_i is its spatial identity (centroid and uncertainty ellipsoid), $\mathcal{E}_i^+$ and $\mathcal{E}_i^-$ hold independent supporting and opposing evidence items, $b_i \in [B_{\min}, B_{\max}]$ is a scalar belief score, and $z_i \in \{\textsc{Tentative}, \textsc{Confirmed}, \textsc{Rejected}\}$ is its lifecycle state, and $n_i^{+,\mathrm{new}}$ counts independent positive groups observed since its latest rejection. CONFIRMED grants approach and benchmark-defined final verification; task success is still determined by the benchmark's stopping rule.

Reference event sets. To avoid defining evaluation success with the method's own internal score, we freeze two external event sets before examining results. $\mathcal{F}_{\mathrm{ref}}$ contains false candidates with a reference contradiction time $t_i^\star$, obtained from simulator identity labels or blinded human annotation. $\mathcal{R}_{\mathrm{ref}}$ contains true-target recovery episodes with a degraded or occluded interval followed by new independent positive views. The sets are disjoint: temporary loss of a true target is not an FCP failure type.

FCP measures. **State-FCP** records whether candidate i stays in the active plan, while **Behavior-FCP** records whether it triggers a concrete action:

$$\mathrm{SFCP}_i = \mathbb{I}\Big[\exists t \geq t_i^\star + \Delta : i \in \mathcal{C}_t^{\mathrm{plan}}\Big], \tag{2}$$
$$\mathrm{BFCP}_i = \mathbb{I}[\exists t \geq t_i^\star + \Delta : a_t \rightsquigarrow i]\,. \tag{3}$$

Here $a_t \rightsquigarrow i$ means that action a_t approaches, revisits, or confirms candidate i. Over $N_F = |\mathcal{F}_{\mathrm{ref}}|$ eligible events, the reported rates are

$$\mathrm{SFCPR} = \frac{1}{N_F} \sum_{i \in \mathcal{F}_{\mathrm{ref}}} \mathrm{SFCP}_i,$$
$$\mathrm{BFCPR} = \frac{1}{N_F} \sum_{i \in \mathcal{F}_{\mathrm{ref}}} \mathrm{BFCP}_i.$$

FCP Waste (distance and time consumed after contradiction): total distance and time attributed to actions $a_t \rightsquigarrow i$ for $t \geq t_i^\star + \Delta$.

Persistence window Δ, source-ID/action-geometry attribution, and annotation rules are fixed before test evaluation. We stratify false candidates into same-category distractors, reflections, repeated texture, and scale/viewpoint confusions; aggregate results use all eligible events.

Recovery measures. For each $j \in \mathcal{R}_{\mathrm{ref}}$, let

$$\mathrm{FR}_j = \mathbb{I}\big[\exists t \in [t_j^-, t_j^+] : z_j^t = \textsc{Rejected}\big]\,,$$
$$\mathrm{OR}_j = \mathbb{I}\Big[\exists t \in (t_j^+, t_j^{\mathrm{end}}] : j \in \mathcal{C}_t^{\mathrm{plan}}\Big]\,.$$

Here $[t_j^-, t_j^+]$ is the frozen degradation interval and t_j^{end} closes the recovery window after new positive views. We report $\mathrm{FRR} = N_R^{-1} \sum_j \mathrm{FR}_j$ and $\mathrm{ORR} = N_R^{-1} \sum_j \mathrm{OR}_j$, where $N_R = |\mathcal{R}_{\mathrm{ref}}|$.

4 Revisable Target Memory

RTM rests on a single interface: every waypoint records the candidate that sourced it, and rejecting that candidate invalidates all of its unexecuted goals. This couples memory revision to planning: the system does not merely update a score, but propagates rejection through the goal queue. RTM implements this interface through two modules (Figure 2). *Candidate Evidence Memory* links each observation to a persistent spatial candidate, groups correlated frames into independent viewpoints, and accumulates signed evidence. *Revise-and-Suppress* maps that evidence to a lifecycle state through a state-dependent transition function and cancels every stale waypoint the moment a candidate is rejected. The planner then reads only candidates that are not rejected.

4.1 Candidate Evidence Memory

Association. Each new observation must be linked to an existing candidate or spawn a new one. RTM computes a weighted association score

$$A(e_t, m_i^t) = w_x A_x + w_f A_f + w_\delta A_\delta, \tag{4}$$

with components

$$A_x = \exp\!\left(-\frac{\|\hat{x}_t - x_i\|_2^2}{\sigma_x^2}\right),$$
$$A_f = \cos\big(f_t, \bar{f}_i^t\big)\,,$$
$$A_\delta = \exp\!\left(-\frac{\Delta t}{\tau_\delta}\right).$$

The three terms measure spatial proximity, appearance similarity, and temporal recency, respectively. Here $\bar{f}_i^t$ is the

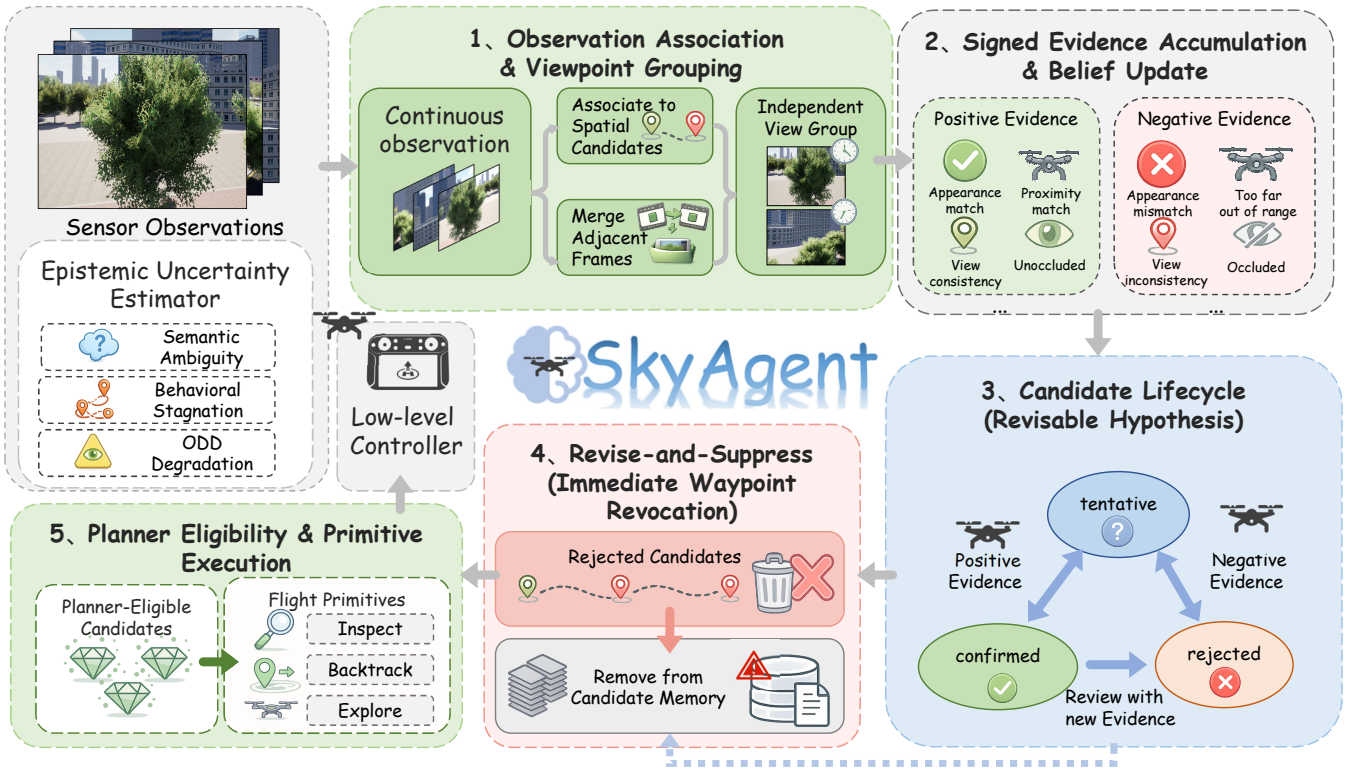


Figure 2: **Revisable Target Memory.** Candidate Evidence Memory groups sequential observations by independent viewpoint and accumulates signed evidence for each persistent candidate. Revise-and-Suppress updates each candidate's lifecycle through a state-dependent transition and propagates rejection to the planner by invalidating every pending waypoint sourced from the rejected candidate.

running mean feature of candidate i, and Δt is the number of steps since i was last updated. The best-scoring candidate is selected: $i^\star = \arg\max_i A(e_t, m_i^t)$. If $A(e_t, m_{i^\star}^t) < \tau_{\text{assoc}}$, a new candidate is created with $b = 0$ and $z = \textsc{Tentative}$. The maximization includes rejected records, which remain in the association and audit store even though they are unavailable to planning. Before creating a new ID, RTM also checks rejected records with the calibrated spatial and appearance gates but without temporal recency; elapsed time alone therefore cannot bypass the reactivation guard. Every create, association, and unassigned event is logged for audit. Failure-source labels are assigned only in the frozen offline evaluation; they are not inputs to association, evidence accumulation, or state transition. An oracle-association ablation (Table 2) separately estimates how much of the remaining error comes from candidate tracking rather than from the revision mechanism itself.

Independent viewpoint groups. Hovering near a false object long enough to collect many similar frames should not raise its confidence score. Two observations belong to the same viewpoint group when all four conditions hold:

$$\begin{aligned} \Delta\theta_{it} &\le \theta_{\max}, & \Delta d_{it} &\le d_{\max}, \\ \Delta h_{it} &\le h_{\max}, & \Delta t &\le T_{\max}. \end{aligned} \tag{5}$$

where $\Delta\theta_{it}$ is the bearing change from p_t toward x_i relative to the group's representative view, Δd_{it} is range change, and Δh_{it} is altitude change. All observations in a group maintain one bounded aggregate contribution; new correlated frames update that aggregate rather than casting additional votes. The counts n_i^+ and n_i^- track the number of distinct groups in $\mathcal{E}_i^+$ and $\mathcal{E}_i^-$ separately; they gate reactivation in Equation 7.

Signed evidence. A frozen query-conditioned match model h_ϕ, calibrated on a held-out split, produces a probability that is converted to the signed evidence y_t defined in the evidence item:

$$r_t = h_\phi(R_t, q), \quad y_t = 2r_t - 1.$$

Here $r_t \in [0, 1]$. Positive values support the candidate's target identity; negative values contradict it; values within $(-\epsilon, \epsilon)$ are discarded as uninformative. The magnitude $|y_t|$ already encodes semantic confidence. For viewpoint group g, define the quality-weighted bounded contribution s_{ig}^t and update the belief by replacing the previous contribution of that group:

$$\begin{aligned} s_{ig}^t &= \frac{\sum_{k \in g^t} \rho_k y_k}{\max(1, \sum_{k \in g^t} \rho_k)}, \\ b_i^{t+1} &= \text{clip}_{[B_{\min}, B_{\max}]}\left(b_i^t + s_{ig}^{t+1} - s_{ig}^t\right). \end{aligned} \tag{6}$$

For a newly created group, $s_{ig}^t = 0$ before its first update. Because $|s_{ig}^t| \le 1$, collecting arbitrarily many correlated

frames cannot increase a group's total evidence mass. Equation 6 is an interpretable evidence accumulator; we do not claim a calibrated Bayesian posterior.

4.2 Revise-and-Suppress

State-dependent lifecycle. Transitions depend on the current score *and* the current state, not on the score alone. Define the four transition guards

$$\begin{aligned}
\gamma_i^{\text{conf}} &\equiv (z_i^t \neq \textsc{Rejected}) \wedge (b_i^{t+1} \geq \tau_{\text{pos}}),\\
\gamma_i^{\text{rej}} &\equiv (z_i^t \neq \textsc{Rejected}) \wedge (b_i^{t+1} \leq \tau_{\text{neg}}),\\
\gamma_i^{\text{dem}} &\equiv (z_i^t = \textsc{Confirmed}) \wedge (b_i^{t+1} < \tau_{\text{pos}} - \delta_{\text{hyst}}),\\
\gamma_i^{\text{react}} &\equiv (z_i^t = \textsc{Rejected}) \wedge (b_i^{t+1} \geq \tau_{\text{react}})\\
&\quad \wedge (n_i^{+,\text{new}} \geq n_{\text{react}}).
\end{aligned}$$

Applied in the order shown below, the lifecycle update is

$$z_i^{t+1} = \begin{cases} \textsc{Confirmed}, & \gamma_i^{\text{conf}},\\ \textsc{Rejected}, & \gamma_i^{\text{rej}},\\ \textsc{Tentative}, & \gamma_i^{\text{dem}} \vee \gamma_i^{\text{react}},\\ z_i^t, & \text{otherwise.} \end{cases} \tag{7}$$

Here $\tau_{\text{neg}} < \tau_{\text{pos}}$ forms the hysteresis band; δ_{hyst} prevents Confirmed from flickering under small score drops; $\tau_{\text{react}} > \tau_{\text{neg}}$ sets a stricter recovery bar; and n_{react} requires at least that many new independent positive viewpoint groups accumulated *after* the most recent rejection. A rejected candidate therefore requires genuinely new independent positive evidence to recover, not a minor score oscillation. On entry to Rejected, RTM resets $n_i^{+,\text{new}}$ to zero. Subsequent independent positive groups increment it once each. Reactivation deliberately returns the candidate only to Tentative; confirmation requires a later update, preventing a single transition from restoring planning authority and confirming the target simultaneously.

Tentative candidates may request additional viewpoints but cannot terminate the episode. Confirmed candidates may be approached and checked under the benchmark's stopping rule. Rejected candidates remain in the audit log but leave the active candidate set; they may return to Tentative only via the reactivation guard in Equation 7.

Planner binding. The planner reads only non-rejected candidates:

$$\mathcal{C}_t^{\text{plan}} = \{m_i^t \mid z_i^t \neq \textsc{Rejected}\}. \tag{8}$$

Every waypoint stores its source candidate ID at generation time. When candidate i enters Rejected, every unexecuted waypoint tagged with ID i is canceled before the next planning cycle begins. An action already under safety-critical execution is completed or terminated by the safety controller rather than interrupted by memory. If eligible candidates remain, planning continues from the remaining set; otherwise the agent resumes coverage search. This cancellation is the step that verify-before-act methods do not take: they replan the next action, but leave previously queued candidate-bound goals intact.

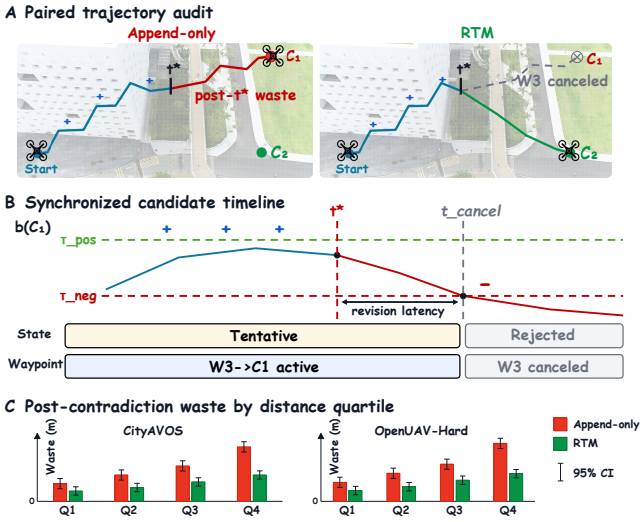


Figure 3: **Mechanism audit layout.** *Panel A* contrasts append-only and RTM trajectories after the reference contradiction at $t^\star$. *Panel B* aligns signed evidence, candidate state, and stale-goal cancellation. *Panel C* defines the distance-quartile summary used for the supplementary stress analysis.

4.3 Integration with Flight Execution

The fast search policy, event trigger, inspection skills, and safety-projected controller are retained as execution infrastructure. They collect observations and execute bounded actions, but candidate state is read and written exclusively through RTM. The language router may emit structured evidence judgments or select `Inspect`; it cannot directly set z_i, reactivate a candidate, or bypass waypoint cancellation. The architecture therefore satisfies a separation of concerns: the revision mechanism can be tested and ablated independently of the specific router or low-level controller.

5 Experiments

We evaluate natural FCP prevalence, controlled benchmark effects, mechanism and recovery, and paired physical transfer.

5.1 Setup

Environments. We use three public aerial benchmarks. **CityAVOS** is used to form 380 Hard tasks and a 380-task Medium control, matched one-to-one by nearest start-to-goal distance (Ji et al. 2026). **UAV-ON** contributes 1,000 test episodes from 14 Unreal Engine environments (Xiao et al. 2025). **OpenUAV-Hard** combines 420 Test-Unseen-Object and 396 Test-Unseen-Map trajectories of at least 250 m under its native protocol (Wang et al. 2025). RTM uses only benchmark-standard observations and goal information.

Protocol. Within each benchmark, RTM variants share task IDs, targets, perception, action budgets, controllers, and evaluators; thresholds, prompts, association rules, and failure handling are fixed on disjoint calibration episodes. Complete-system baselines use official implementations when avail-

Benchmark	Method	SR↑	Eff.↑	Dist.↓	BFCPR↓	Waste↓	FRR↓
CityAVOS-Hard (SkyAgent n=380)	RE (Ji et al. 2026)	5.05	2.47	674.16	N/A	N/A	N/A
	FBE (Ji et al. 2026)	8.07	6.34	628.70	N/A	N/A	N/A
	STMR (Ji et al. 2026)	20.96	13.23	547.77	N/A	N/A	N/A
	PRPSearcher (Hard subset) (Ji et al. 2026)	29.62	16.65	507.60	N/A	N/A	N/A
	Append-only SkyAgent	33.95	23.16	431.13	45.18	120.76	N/A
	SkyAgent + RTM (Hard)	**40.26**	**30.14**	**349.14**	**15.27**	**44.21**	**2.91**
CityAVOS-Medium (SkyAgent n=380)	PRPSearcher (Medium subset) (Ji et al. 2026)	42.89	36.68	362.11	N/A	N/A	N/A
	Append-only SkyAgent (Medium)	45.26	36.85	360.48	20.39	55.23	N/A
	SkyAgent + RTM (Medium)	**48.16**	**40.11**	**329.13**	**7.69**	**24.82**	**1.94**
UAV-ON (n=1,000)	Random (Zhou et al. 2026)	3.70	2.66	596.53	N/A	N/A	N/A
	CLIP-H (Zhou et al. 2026)	6.20	4.15	547.32	N/A	N/A	N/A
	AOA-F (Zhou et al. 2026)	7.30	4.06	550.29	N/A	N/A	N/A
	OctMem-Agent (Zhou et al. 2026)	19.50	6.37	474.00	N/A	N/A	N/A
	Append-only SkyAgent	20.30	6.53	468.72	29.95	90.97	N/A
	SkyAgent + RTM	**23.20**	**8.07**	**417.86**	**12.32**	**40.71**	**3.09**
OpenUAV-Hard (n=816)	Random Action (Zhang et al. 2026b)	0.00	0.00	800.66	N/A	N/A	N/A
	Fixed Action (Zhang et al. 2026b)	1.10	0.71	776.22	N/A	N/A	N/A
	CMA (Zhang et al. 2026b)	3.43	3.37	684.63	N/A	N/A	N/A
	TravelUAV (Zhang et al. 2026b)	12.74	11.76	395.77	N/A	N/A	N/A
	Append-only SkyAgent	18.01	7.03	558.62	35.32	120.90	N/A
	SkyAgent + RTM	**22.06**	**9.03**	**489.76**	**14.07**	**50.41**	**3.52**

Table 1: **Performance on aerial benchmarks.** SkyAgent rows are paired internal comparisons. External SR and Eff. values are reported by the cited papers; Dist. values are outputs of our benchmark reproduction rather than values copied from their tables. External methods may use different perception front ends. SR, Eff., BFCPR, and FRR are percentages; Dist. and Waste are meters.

able. SPF and PSC-AVDN are adapted to the common DJI interface without task-specific training; their table values are measured under our protocol. Success uses an exact paired binary test; continuous and event measures use paired, episode-clustered bootstrap intervals.

Metrics. We report benchmark-native SR, path efficiency, distance, and time. Mechanism metrics are SFCPR, BFCPR, post-contradiction Waste, and revision latency. FRR and ORR measure suppression cost and recovery; association accuracy and model calls are supporting diagnostics.

5.2 FCP in Unmodified Search

Among 380 append-only CityAVOS-Hard episodes, 197 contain a reference event; 89 contradicted candidates remain active (BFCPR 45.18%), causing 120.76 m of post-contradiction travel. Two annotators label creation, the first reference contradiction, later eligibility, false confirmation, and attributable distance under fixed $t_i^\star$ rules, with adjudication across five distractor, appearance, viewpoint, occlusion, and degradation strata.

5.3 Controlled Benchmark Results

FCP suppression. Across four splits, RTM reduces CityAVOS-Hard BFCPR from 45.18% to 15.27% (29.91 points; 66.2% relative) and reduces post-contradiction waste from 120.76 m to 44.21 m. The same pattern holds on CityAVOS-Medium (12.70 points and 30.41 m), UAV-ON (17.63 points and 50.26 m), and OpenUAV-Hard (21.25 points and 70.49 m), while FRR remains 1.94–3.52%.

Task outcomes. RTM improves SR and path efficiency on every split; SR gains are 2.90–6.31 points and peak on ambiguity-rich CityAVOS-Hard. SkyAgent comparisons use identical paired episodes and complete splits. CityAVOS baseline values come from the difficulty-stratified 605-task preprint evaluation (Ji et al. 2026); unavailable stratum counts are therefore not assigned to those rows. OpenUAV-Hard baselines combine the published Test-Unseen-Object (n=420) and Test-Unseen-Map (n=396) Hard aggregates. Eff. is paper-reported SPL and N/A marks unavailable RTM diagnostics. BFCPR and Waste reductions support revision as the mechanism, although route changes also shift downstream observations.

5.4 Mechanism, Recovery, and Stress

Component isolation. Rechecking only the current action leaves BFCPR at 38.19% and Waste at 96.17 m, versus 15.27% and 44.21 m for Full RTM. State revision without queued-goal cancellation reaches 28.86% BFCPR and 74.24 m Waste, isolating the value of planner binding. Removing negative evidence approaches append-only behavior; removing viewpoint grouping raises FRR from 2.91% to 8.65%.

Permanent rejection lowers BFCPR but raises FRR to 11.65% and reduces ORR to 65.63%; Full RTM preserves 92.71% ORR with 2.91% FRR. Oracle association reaches 42.11% SR and 8.82% BFCPR, identifying cross-view association as the main remaining bottleneck.

Long-horizon and recovery controls. On 816 OpenUAV-Hard trajectories, RTM lowers BFCPR from 35.32% to

Variant	SR↑	SFCPR↓	BFCPR↓	Waste↓	FRR↓	ORR↑	Latency↓
Append-only	33.95	55.33	45.18	120.76	–	77.08	–
Verify current action	36.05	50.25	38.19	96.17	1.96	80.21	6.53
RTM w/o negative evidence	35.53	51.24	39.80	103.18	0.99	82.29	7.29
RTM w/o viewpoint grouping	37.63	27.23	21.29	60.92	8.65	77.08	2.37
Revision state only	38.42	19.90	28.86	74.24	3.88	89.58	8.72
Permanent rejection	38.68	15.42	12.44	38.26	11.65	65.63	2.52
Full RTM	**40.26**	**16.75**	**15.27**	**44.21**	**2.91**	**92.71**	**3.19**
Full + oracle association	42.11	9.80	8.82	29.50	1.92	95.83	2.54

Table 2: **Mechanism ablations on CityAVOS-Hard** (n=380)**.** Closed-loop rates use row-specific externally annotated events; the shared-stream analysis and all event counts are reported separately. Waste is meters and Latency is steps.

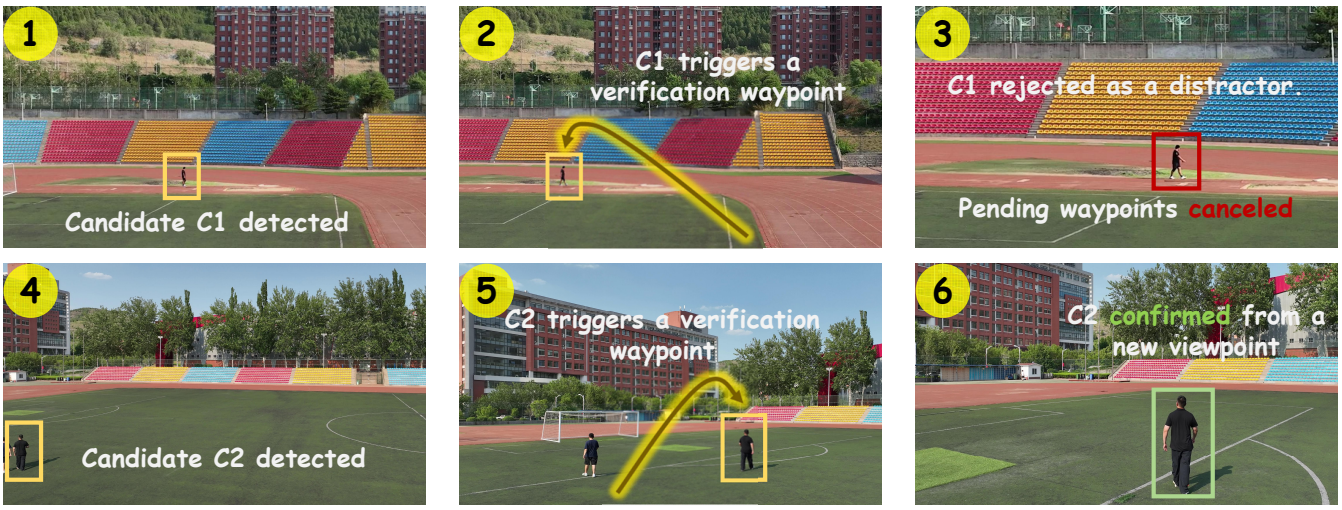


Figure 4: **Representative real-flight revision sequence.** Inspection rejects C_1 and cancels its queued waypoint; search resumes toward C_2, confirmed from an independent view. Table 3 reports all paired trials.

14.07% and Waste from 120.90 m to 50.41 m while SR rises from 18.01% to 22.06%. The CityAVOS-Hard control confirms the FRR/ORR trade-off; Figure 3 audits trajectories, candidate states, and cancellation, with shared-stream and stratified analyses in the supplement.

5.5 Real-Flight Evaluation

Method	SR↑	Dist.↓	BFCPR↓	FRR↓	Rev.Lat.(s)↓	Int.↓
SPF (TF)	54.29	275.68	50.00	12.50	6.31	17.14
PSC-AVDN (TF)	60.00	257.94	40.91	16.67	5.24	11.43
Append-only SkyAgent	57.14	264.37	45.45	–	–	14.29
SkyAgent + RTM	**71.43**	**223.81**	**13.64**	**4.17**	**2.78**	**5.71**

Table 3: **Matched DJI flights.** Results use 35 matched scenarios/method (140 trials). Rates are percentages, Dist. is meters, and Rev.Lat. is seconds; TF denotes no task-specific training or fine-tuning.

With the common backend, training-free SPF and PSC-AVDN reach 54.29/60.00% SR and 50.00/40.91% BFCPR (Hu et al. 2025; Qi et al. 2026). Against append-only, RTM raises SR from 57.14% to 71.43%, cuts distance from 264.37 m to 223.81 m, BFCPR from 45.45% to 13.64%, and interventions from 14.29% to 5.71%; FRR is 4.17% and latency 2.78 s. The four methods share 35 starts, regions, layouts, weather strata, and budgets; order is randomized and all outcomes retained across five conditions. Figure 4 shows one sequence; Table 3 covers 140 trials.

6 Limitations and Conclusion

RTM assumes reliable cross-view association; merging or splitting candidates distorts FCP counts, and oracle association estimates this tracking contribution. The frozen evidence model may retain systematic biases. Because RTM also changes later routes and observations, closed-loop results measure system behavior; common-trajectory replay is needed to isolate revision. CityAVOS-calibrated thresholds may require recalibration for new cameras or altitudes. The present scope covers static objects, and the tested flights do not establish a general safety guarantee.

Candidate Evidence Memory aggregates signed evidence across viewpoints, while Revise-and-Suppress cancels waypoints sourced by rejected candidates. RTM improves task performance across four paired benchmark splits while reducing FCP and keeping false rejection low.

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